Correcao Prova

1a) $v(t) = 6t - 3t^2$ } derivadas
$a(t) = 6 - 6t$

2)

b) $\frac{d\,x(t)}{dt} = 0$

$v(t) = 0$

$6t - 3t^2 = 0$

$t(6 - 3t) = 0$

$t = 0s$

$\boxed{t_2 = 2s}$

c) $V_m = \frac{\Delta x}{\Delta t}$

$t(2,5)$

$V_m(2,5) = \frac{x(2,5) - x(0)}{2,5} = \frac{3.125 - 0}{2,5} = \boxed{1,25 m/s}$

$x(2,5) = 3t^2 - 1t^3$

$18,75 - 15,625$

$\boxed{3,125 m}$

$x(0) = 0$

d) $v(2,5) = 6.2,5 - 3.2,5^2$

$15 - 18,75$

$\boxed{-3,75 m/s}$

e) $a_m = \frac{\Delta v}{\Delta t}$

$\frac{v(2,5) - V(0)}{2,5} \quad \frac{-3,75}{2,5} = \boxed{-1,5 m/s^2}$

f) $a(2,5) = 6 - 6.2,5$

$= \boxed{-9 m/s^2}$

g) Dist $= 4 + (4 - 3,125)$

Deslocamento ≠ dist

$\boxed{\text{Dist} = 4,875 m}$